

QUIZ 5

Instructions: Please write your name at the top of the page. A precise statement in each cell is enough; no proofs are required.

(a) (10 points). Let $\mathbb{K}$ be a field. Let V, V', W be finite-dimensional $\mathbb{K}$ -vector spaces, with bases $e_1, \dots, e_n$ of V and $f_1, \dots, f_m$ of W . In Table (a), let $\phi: V \rightarrow V'$ be linear. Complete both tables.

- For $T(V)$, $\Lambda(V)$, and $\text{Sym}(V)$, give the dimension and a basis of the degree- d piece ($d \geq 0$), and the dimension of the whole algebra.
- For each induced map, state its domain, codomain, and action on elements or generators. For $V \otimes W$ and $\text{Hom}_{\mathbb{K}}(V, W)$, keep W fixed.

Table (a). Dimensions, bases, and induced maps.

Object	Dimension	A basis	Map induced by ϕ
$V \otimes W$			
$V^\vee$			
$\text{Hom}_{\mathbb{K}}(V, W)$			
$T(V)$			
$\Lambda(V)$			
$\text{Sym}(V)$			

Table (b). Constructions and universal properties.

Name the canonical map in each universal property, and state the corresponding unique factorization or bijection of maps. In the last column, give *one* adjunction or natural isomorphism.

Here U may be any $\mathbb{K}$ -vector space. All algebras and algebra homomorphisms are unital; all maps are over $\mathbb{K}$. State any additional conditions on a target algebra A .

Object	Definition or construction	Universal property	An adjunction or natural isomorphism
$V \otimes W$			
$V^\vee$			
$\text{Hom}_{\mathbb{K}}(V, W)$			
$T(V)$			
$\Lambda(V)$			
$\text{Sym}(V)$			